BLUE TEAM WEEK 2 INTERNSHIP REPORT

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DATE: 3/15/2026

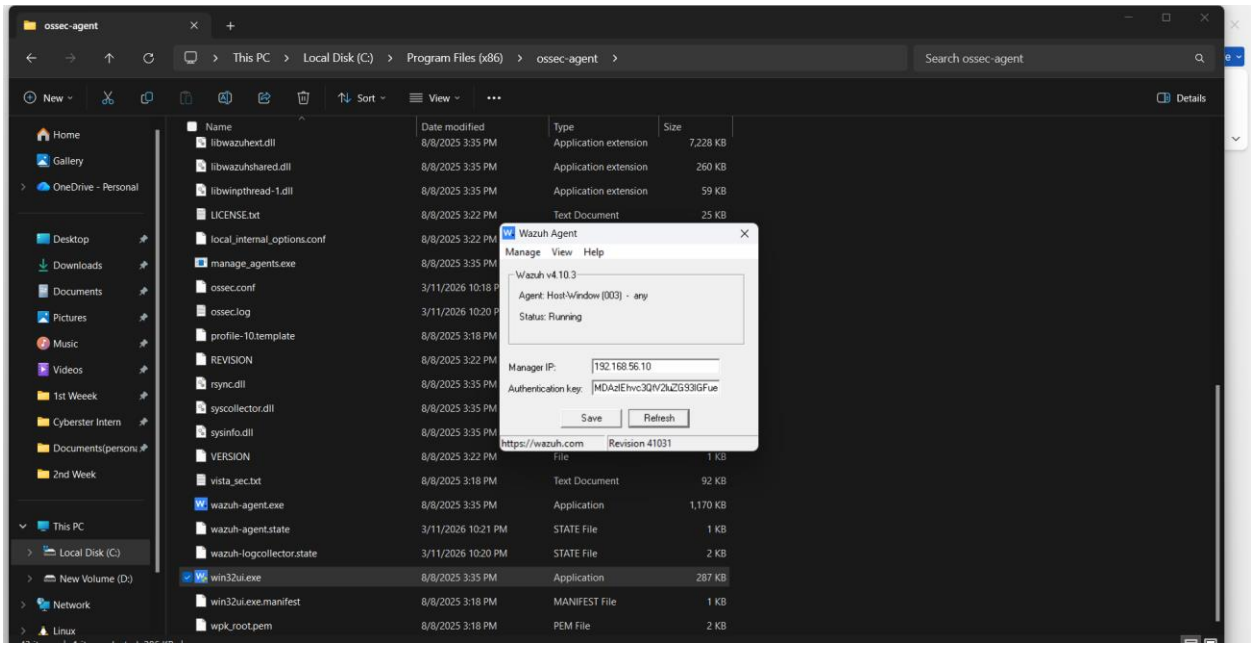
REPORT WEEK 2

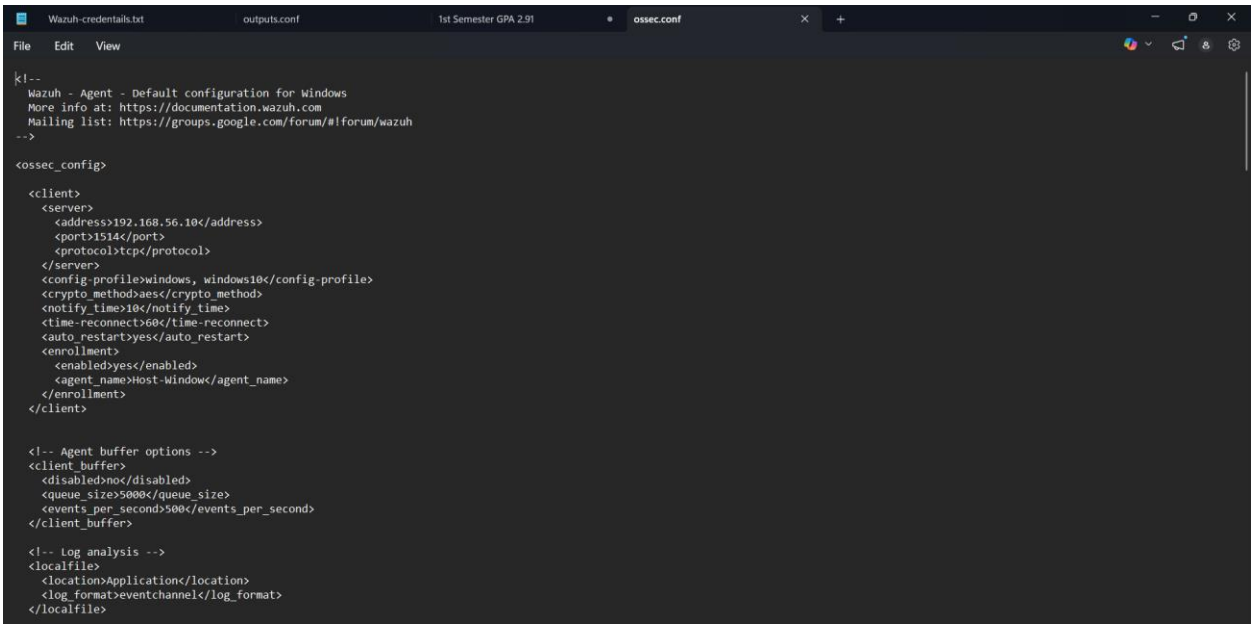
2nd Week

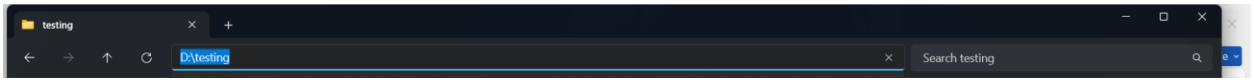
Days 8 and 9 — File Integrity Monitoring

Deliverables:

- A detailed report covering every step of the FIM configuration. Start from the ossec.conf changes and walk through to the alerts appearing in the dashboard. Include screenshots at each stage, not just the final result.

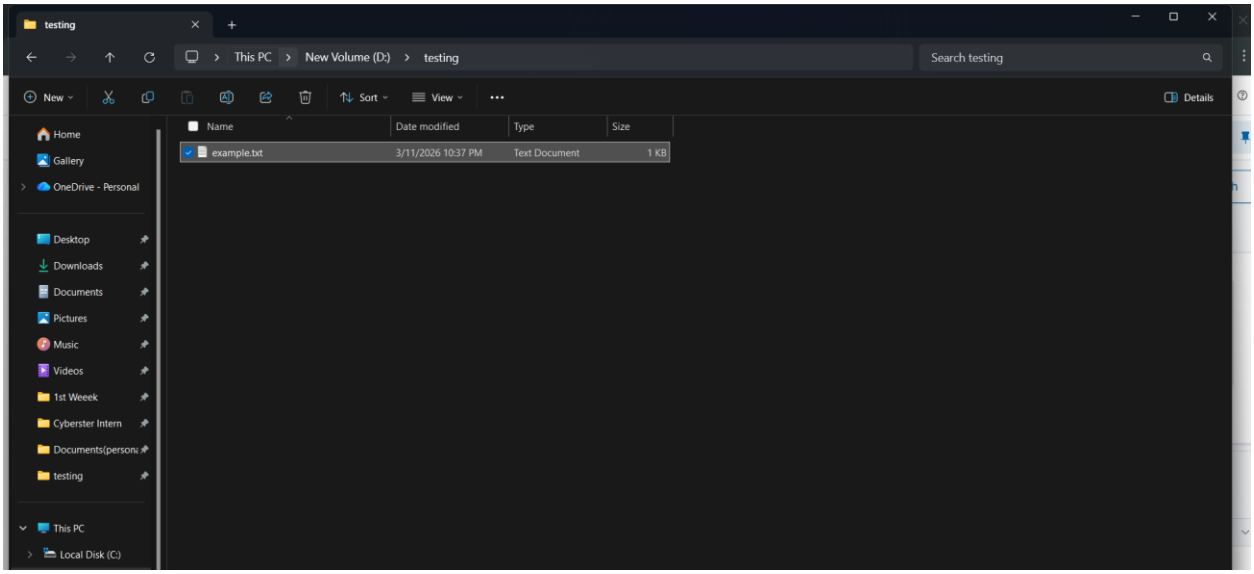


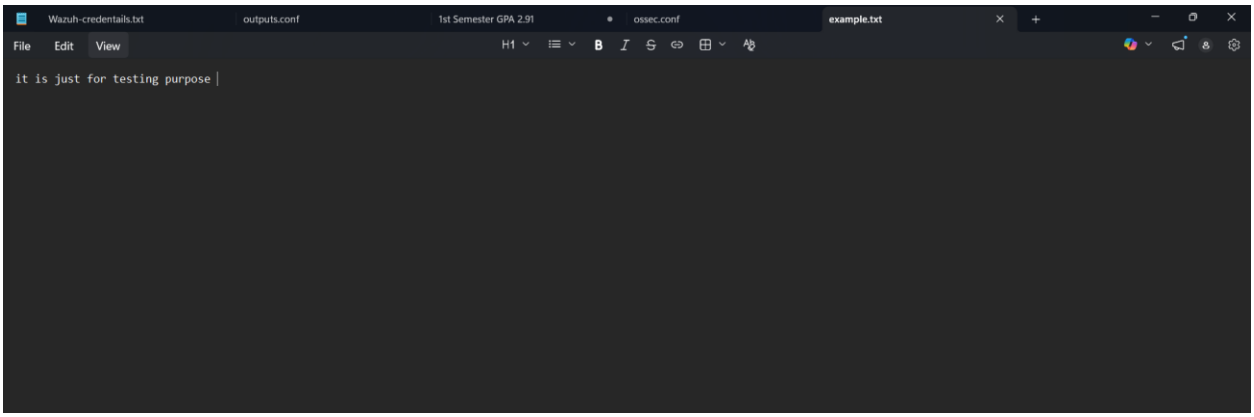




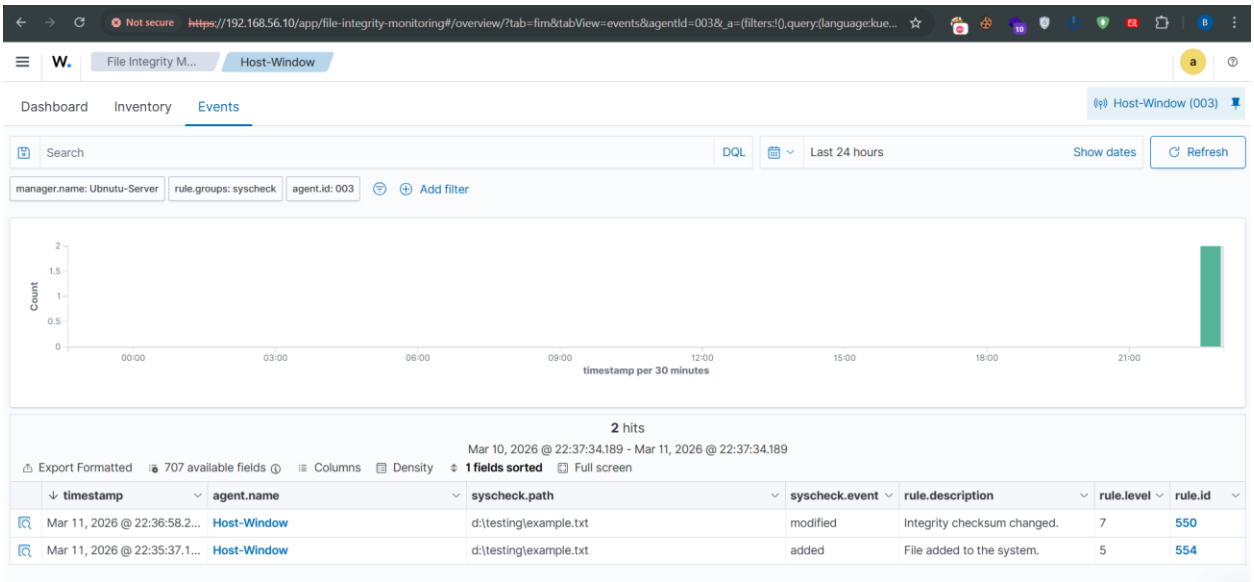
REPORT WEEK 2

```
  <!-- File integrity monitoring -->
  <syscheck>

    <disabled>no</disabled>

    <!-- Frequency that syscheck is executed default every 12 hours -->
    <frequency>43200</frequency>

    <!-- Default files to be monitored. -->
    <directories recursion_level="0" restrict="regedit.exe$|system.ini$|win.ini$">%WINDIR%</directories>

    <directories recursion_level="0" restrict="at.exe$|attrib.exe$|cacls.exe$|cmd.exe$|eventcreate.exe$|ftp.exe$|lsass.exe$|net.exe$|net1.exe$|netsh.exe$|reg.exe
$|regedt32.exe|regsvr32.exe|runas.exe|sc.exe|schtasks.exe|sethc.exe|subst.exe$">%WINDIR%\SysNative</directories>
    <directories recursion_level="0">%WINDIR%\SysNative\drivers\etc</directories>
    <directories recursion_level="0" restrict="WMIC.exe$">%WINDIR%\SysNative\wbem</directories>
    <directories recursion_level="0" restrict="powershell.exe$">%WINDIR%\SysNative\WindowsPowerShell\v1.0</directories>
    <directories recursion_level="0" restrict="winrm.vbs$">%WINDIR%\SysNative</directories>


    <directories check_all="yes" realtime="yes" report_changes="yes">D:\testing</directories>
```





REPORT WEEK 2



- The ossec.conf syscheck block from one of your agents, showing your full FIM configuration.

```
  <!-- File integrity monitoring -->
  <syscheck>

    <disabled>no</disabled>

    <!-- Frequency that syscheck is executed default every 12 hours -->
    <frequency>43200</frequency>

    <!-- Default files to be monitored. -->
    <directories recursion_level="0" restrict="regedit.exe$|system.ini$|win.ini$">%WINDIR%</directories>

    <directories recursion_level="0" restrict="at.exe$|attrib.exe$|cacls.exe$|cmd.exe$|eventcreate.exe$|ftp.exe$|lsass.exe$|net.exe$|net1.exe$|netsh.exe$|reg.exe
$|regedt32.exe|regsvr32.exe|runas.exe|sc.exe|schtasks.exe|sethc.exe|subst.exe$">%WINDIR%\SysNative</directories>
    <directories recursion_level="0">%WINDIR%\SysNative\drivers\etc</directories>
    <directories recursion_level="0" restrict="WMIC.exe$">%WINDIR%\SysNative\wbem</directories>
    <directories recursion_level="0" restrict="powershell.exe$">%WINDIR%\SysNative\WindowsPowerShell\v1.0</directories>
    <directories recursion_level="0" restrict="winrm.vbs$">%WINDIR%\SysNative</directories>


    <directories check_all="yes" realtime="yes" report_changes="yes">D:\testing</directories>
```

- A FIM event analysis report of one to two pages. For each test event you generated, document the file path, the type of change, the timestamp, the agent, and the user. Write a short paragraph below the table explaining what you would do next if this change had appeared in a real environment and you did not know who made it
- {
- "_index": "wazuh-alerts-4.x-2026.03.11",
- "_id": "iW753ZwBVPigi-hNPl_q",
- "_version": 1,
- "_score": null,

```
"_source": {
  "syscheck": {
    "size_before": "14",
    "uname_after": "Mr-NooB",
    "mtime_after": "2026-03-11T17:37:05",
    "size_after": "31",
    "md5_before": "fa47bb06f6ea83e49cc70cd2a130f1dc",
    "diff": "< happy birthday\n---\n> it is just for testing purpose \n",
    "win_perm_after": [
      {
        "allowed": [
          "DELETE",
          "READ_CONTROL",
          "WRITE_DAC",
          "WRITE_OWNER",
          "SYNCHRONIZE",
          "READ_DATA",
          "WRITE_DATA",
          "APPEND_DATA",
          "READ_EA",
          "WRITE_EA",
          "EXECUTE",
          "READ_ATTRIBUTES",
          "WRITE_ATTRIBUTES",
          [
            "DELETE",
            "READ_CONTROL",
            "WRITE_DAC",
            "WRITE_OWNER",
            "SYNCHRONIZE",
            "READ_DATA",
            "WRITE_DATA",
            "APPEND_DATA",
            "READ_EA",
            "WRITE_EA",
            "EXECUTE",
            "READ_ATTRIBUTES",
            "WRITE_ATTRIBUTES"
          ],
          [
```

```
            "DELETE",
            "READ_CONTROL",
            "SYNCHRONIZE",
            "READ_DATA",
            "WRITE_DATA",
            "APPEND_DATA",
            "READ_EA",
            "WRITE_EA",
            "EXECUTE",
            "READ_ATTRIBUTES",
            "WRITE_ATTRIBUTES"
          ],
          [
            "READ_CONTROL",
            "SYNCHRONIZE",
            "READ_DATA",
            "READ_EA",
            "EXECUTE",
            "READ_ATTRIBUTES"
          ]
        ],
        "name": "Administrators"
      },
      {
        "allowed": [
          "DELETE",
          "READ_CONTROL",
          "WRITE_DAC",
          "WRITE_OWNER",
          "SYNCHRONIZE",
          "READ_DATA",
          "WRITE_DATA",
          "APPEND_DATA",
          "READ_EA",
          "WRITE_EA",
          "EXECUTE",
          "READ_ATTRIBUTES",
          "WRITE_ATTRIBUTES"
        ],
        "name": "SYSTEM"
```

```
        },
        {
          "allowed": [
            "DELETE",
            "READ_CONTROL",
            "SYNCHRONIZE",
            "READ_DATA",
            "WRITE_DATA",
            "APPEND_DATA",
            "READ_EA",
            "WRITE_EA",
            "EXECUTE",
            "READ_ATTRIBUTES",
            "WRITE_ATTRIBUTES"
          ],
          "name": "Authenticated Users"
        },
        {
          "allowed": [
            "READ_CONTROL",
            "SYNCHRONIZE",
            "READ_DATA",
            "READ_EA",
            "EXECUTE",
            "READ_ATTRIBUTES"
          ],
          "name": "Users"
        }
      ],
      "sha256_before": "d7469a66c4bb97c09aa84e8536a85f1795761f5fe01ddc8139922b6236f4397d",
      "mtime_before": "2026-03-11T17:35:44",
      "mode": "realtime",
      "path": "d:\\testing\\example.txt",
      "sha1_after": "7a83b27e126aef353fae195eb4341be68611dc23",
      "changed_attributes": [
        "size",
        "mtime",
        "md5",
        "sha1",
```

```
        "sha256"
      ],
      "attrs_after": [
        "ARCHIVE"
      ],
      "uid_after": "S-1-5-21-1579562968-2063753537-1316154528-1001",
      "event": "modified",
      "md5_after": "afa608809e2d2af1c8b5ad7735372cb9",
      "sha1_before": "d324d4eb16b747f383c5e9c2598d209a6b6efad1",
      "sha256_after":
"dd0a409784c942fef8fd2adf7669c13545942cc7f05079164a729952b057a4bd"
    },
    "input": {
      "type": "log"
    },
    "agent": {
      "ip": "192.168.56.101",
      "name": "Host-Window",
      "id": "003"
    },
    "manager": {
      "name": "Ubnutu-Server"
    },
    "rule": {
      "mail": false,
      "level": 7,
      "pci_dss": [
        "11.5"
      ],
      "hipaa": [
        "164.312.c.1",
        "164.312.c.2"
      ],
      "tsc": [
        "PI1.4",
        "PI1.5",
        "CC6.1",
        "CC6.8",
        "CC7.2",
        "CC7.3"
```

```
      ],
      "description": "Integrity checksum changed.",
      "groups": [
        "ossec",
        "syscheck",
        "syscheck_entry_modified",
        "syscheck_file"
      ],
      "nist_800_53": [
        "SI.7"
      ],
      "gdpr": [
        "II_5.1.f"
      ],
      "firedtimes": 1,
      "mitre": {
        "technique": [
          "Stored Data Manipulation"
        ],
        "id": [
          "T1565.001"
        ],
        "tactic": [
          "Impact"
        ]
      },
      "id": "550",
      "gpg13": [
        "4.11"
      ]
    },
    "location": "syscheck",
    "decoder": {
      "name": "syscheck_integrity_changed"
    },
    "id": "1773250618.2221663",
    "full_log": "File 'd:\\testing\\example.txt' modified\nMode: realtime\nChanged attributes: size,mtime,md5,sha1,sha256\nSize changed from '14' to '31'\nOld modification time was: '1773250544', now it is '1773250625'\nOld md5sum was: 'fa47bb06f6ea83e49cc70cd2a130f1dc'\nNew md5sum is :
```

'afa608809e2d2af1c8b5ad7735372cb9'\nOld sha1sum was: 'd324d4eb16b747f383c5e9c2598d209a6b6efad1'\nNew sha1sum is : '7a83b27e126aef353fae195eb4341be68611dc23'\nOld sha256sum was: 'd7469a66c4bb97c09aa84e8536a85f1795761f5fe01ddc8139922b6236f4397d'\nNew sha256sum is : 'dd0a409784c942fef8fd2adf7669c13545942cc7f05079164a729952b057a4bd'\n",
- "timestamp": "2026-03-11T17:36:58.203+0000"
- },
- "fields": {
- "syscheck.mtime_after": [
- "2026-03-11T17:37:05.000Z"
-],
- "syscheck.mtime_before": [
- "2026-03-11T17:35:44.000Z"
-],
- "timestamp": [
- "2026-03-11T17:36:58.203Z"
-]
- },
- "sort": [
- 1773250618203
-]
- }

After getting this alert from real environment, then my first approach to should verify this event and see what type of changed happened, it is delete, update and added

Days 10 and 11 — Custom Wazuh Rules

Deliverables:

- Your custom rules XML file containing all three rules. The file should be clean, commented to explain what each rule does, and structured so someone else could read and maintain it.

```
  </rule>

  <rule id="100002" level="10">
    <if_sid>5501</if_sid>
    <description>Custom Alert: New Linux user account created</description>
    <group>account_management</group>
    <mitre>
      <id>T1136</id>
    </mitre>
   </rule>

   <rule id="100003" level="12" frequency="2" timeframe="120">
     <if_sid>5716</if_sid>
     <same_source_ip />
     <description>Possible SSH brute force attack detected</description>
     <group>authentication_failure, brute_force</group>
     <mitre>
       <sid>T1110</sid>
     </mitre>
    </rule>

  <rule id="100004" level="8">
    <field name="win.system.eventID">2003</field>
    <description> USB storage device inserted on Windows system</description>
    <group>device, usb</group>
    <mitre>
      <id>T1091</id>
    </mitre>
  </rule>


</group>
```

- Screenshots of each rule firing in the Wazuh dashboard, showing the alert description, the level, and the agent it came from.

Rule id=100002

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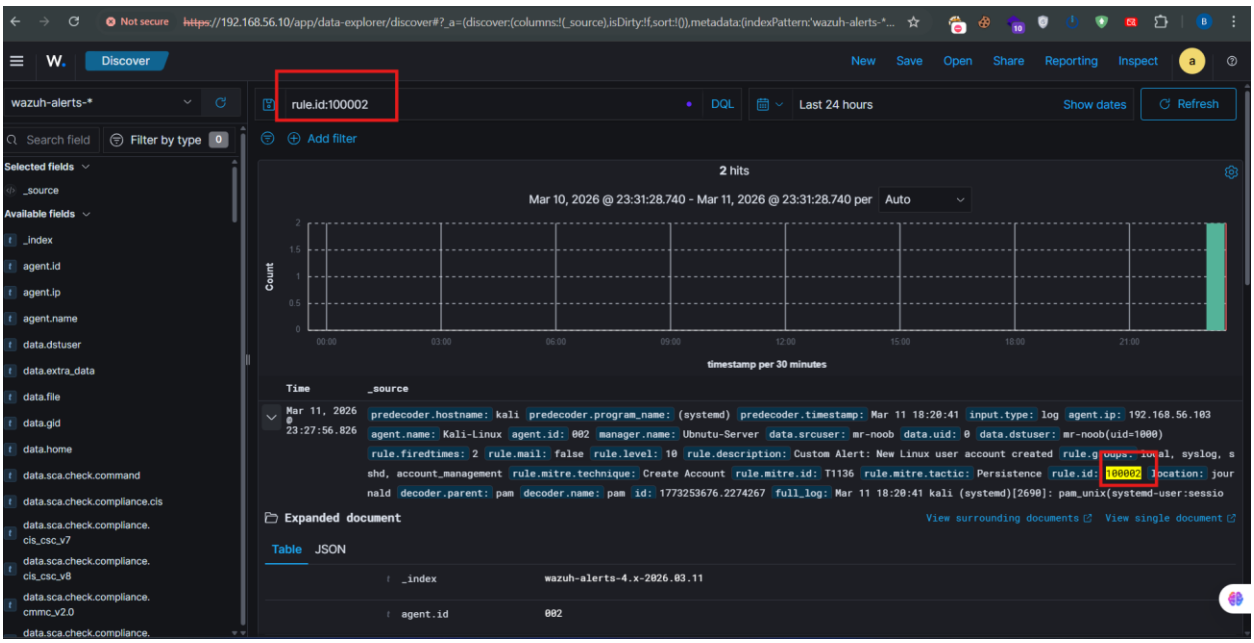
```
mr-noob@kali: ~
┌──(mr-noob㉿kali)-[~]
└─$ sudo adduser testuser1
[sudo] password for mr-noob:
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for testuser1
Enter the new value, or press ENTER for the default
        Full Name []:
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] y

┌──(mr-noob㉿kali)-[~]
└─$
```



Rule id=100003

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Mr-NooB> ssh mr-noob@192.168.56.103
ssh: connect to host 192.168.56.103 port 22: Connection refused
PS C:\Users\Mr-NooB> ssh mr-noob@192.168.56.103
mr-noob@192.168.56.103's password:
Permission denied, please try again.
mr-noob@192.168.56.103's password:
Permission denied, please try again.
mr-noob@192.168.56.103's password:
mr-noob@192.168.56.103: Permission denied (publickey,password).
PS C:\Users\Mr-NooB>
```

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predecoder.timestamp	Mar 11 18:34:23
rule.description	Possible SSH brute force attack detected
rule.firedtimes	3
rule.groups	local, syslog, sshd, authentication_failure, brute_force
rule.id	100003
rule.level	12
rule.mail	true
rule.mitre.id	T1110
rule.mitre.tactic	Credential Access
rule.mitre.technique	Brute Force
timestamp	Mar 11, 2026 @ 23:34:18.676

- A rule tuning notes document. For each rule, write a short paragraph describing what the rule detects, what condition you tested it with, whether it produced any false positives, and what you changed to address them. This is the kind of documentation a real detection engineering team keeps so they know the history of a rule.

Rule id=100002— New User Creation

This rule detects creation of a new Linux user account. The rule builds on Wazuh base rule 5501. I tested it by running sudo useradd attacker on the Ubuntu agent. The alert appeared correctly in the Wazuh dashboard with level 10 severity. No false positives were observed.

Rule id=100003 — SSH Brute Force

This rule detects repeated SSH authentication failures from the same IP. The threshold is five failures within 120 seconds. I simulated the attack from Kali Linux by repeatedly attempting SSH login with an incorrect password. Initially the rule triggered too early, so the frequency threshold was adjusted to five attempts.

Rule id=100004 — USB Device Insertion

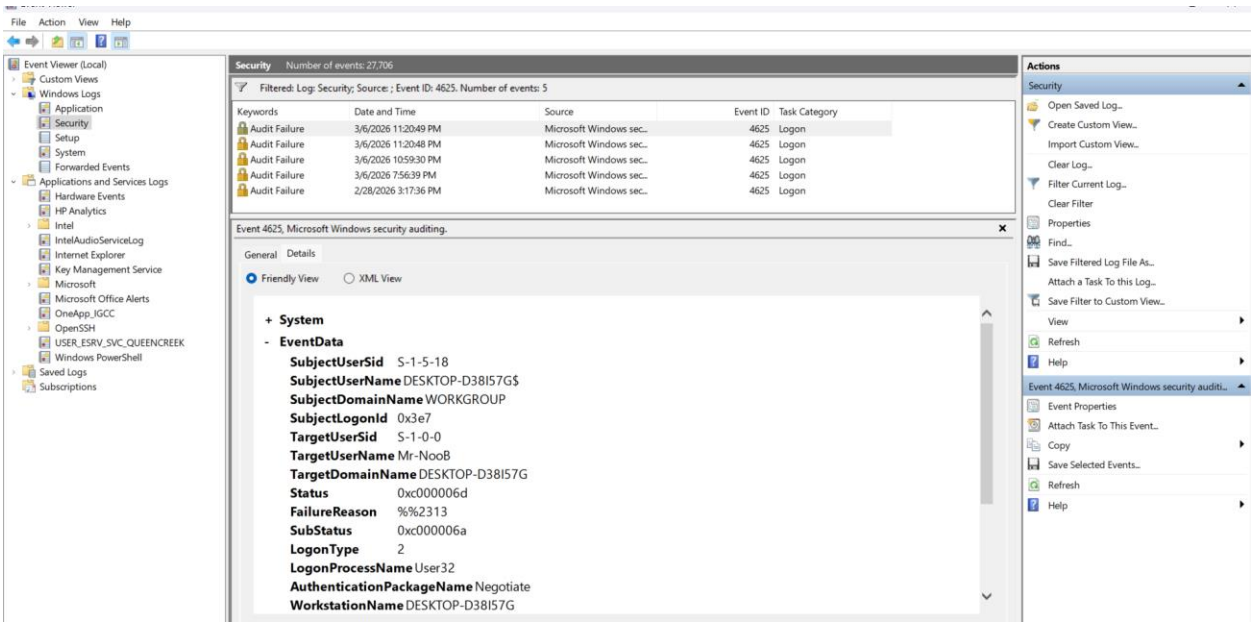
This rule detects Windows event ID 2003 indicating USB device insertion. The test was performed by connecting a USB drive to the Windows VM. The alert appeared correctly in the Wazuh dashboard.

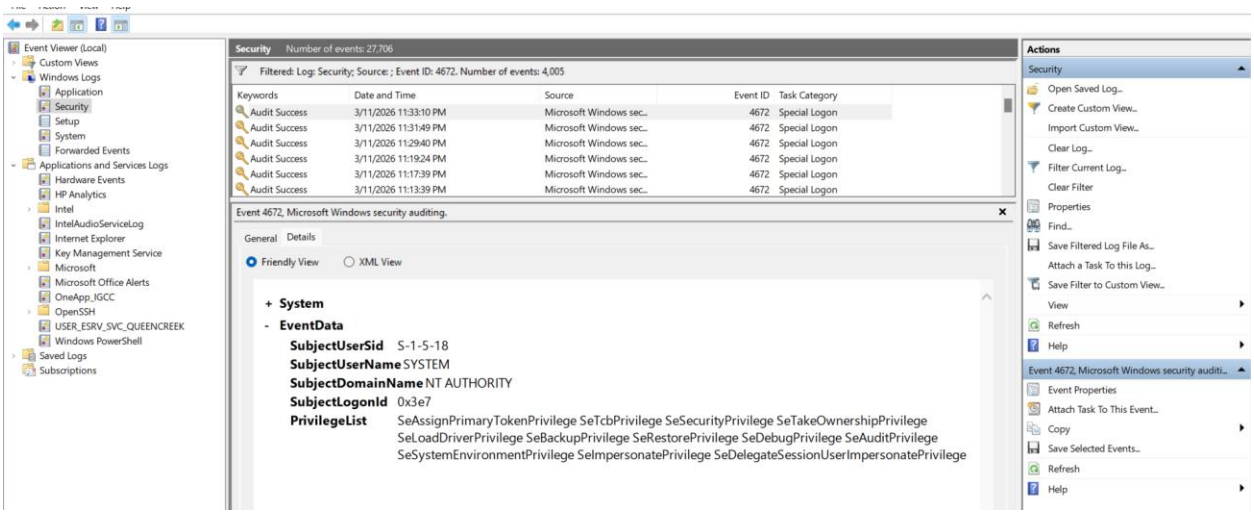
Days 12 and 13 — Log Analysis Fundamentals

Deliverables:

- A log analysis report covering your findings from both Linux and Windows sources. Structure it by source, then by event type. For each finding, include the raw event detail, your interpretation of what it represents, and whether you believe it warrants further investigation or is routine activity.

```
┌──(mr-noob㉿kali)-[/var/log]
└─$ tail auth.log
2026-03-11T13:34:19.270803-05:00 kali sshd-session[9205]: Failed password for mr-noob from 192.168.56.101 port 62432 ssh2
2026-03-11T13:34:21.085981-05:00 kali unix_chkpwd[9263]: password check failed for user (mr-noob)
2026-03-11T13:34:23.198932-05:00 kali sshd-session[9205]: Failed password for mr-noob from 192.168.56.101 port 62432 ssh2
2026-03-11T13:34:23.360843-05:00 kali sshd-session[9205]: Connection reset by authenticating user mr-noob 192.168.56.101 port 62432 [preauth]
2026-03-11T13:34:23.361212-05:00 kali sshd-session[9205]: PAM 2 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.56.101  user=mr-noob
2026-03-11T13:35:01.771090-05:00 kali CRON[9507]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-03-11T13:35:01.793743-05:00 kali CRON[9507]: pam_unix(cron:session): session closed for user root
2026-03-11T13:39:01.825797-05:00 kali CRON[10974]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-03-11T13:39:01.845458-05:00 kali CRON[10974]: pam_unix(cron:session): session closed for user root
2026-03-11T13:42:17.614760-05:00 kali gdm-password]: gkr-pam: unlocked login keyring
```





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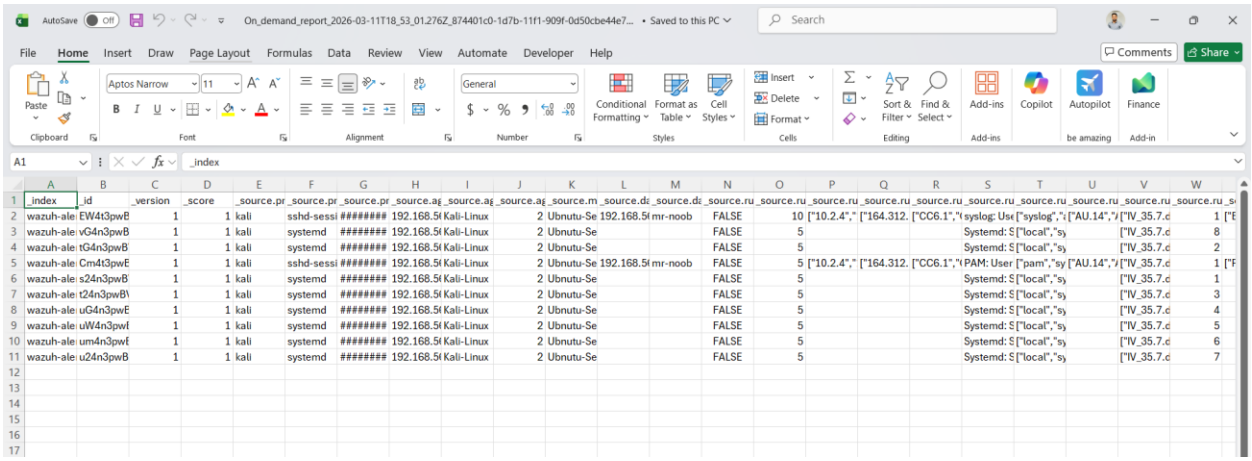
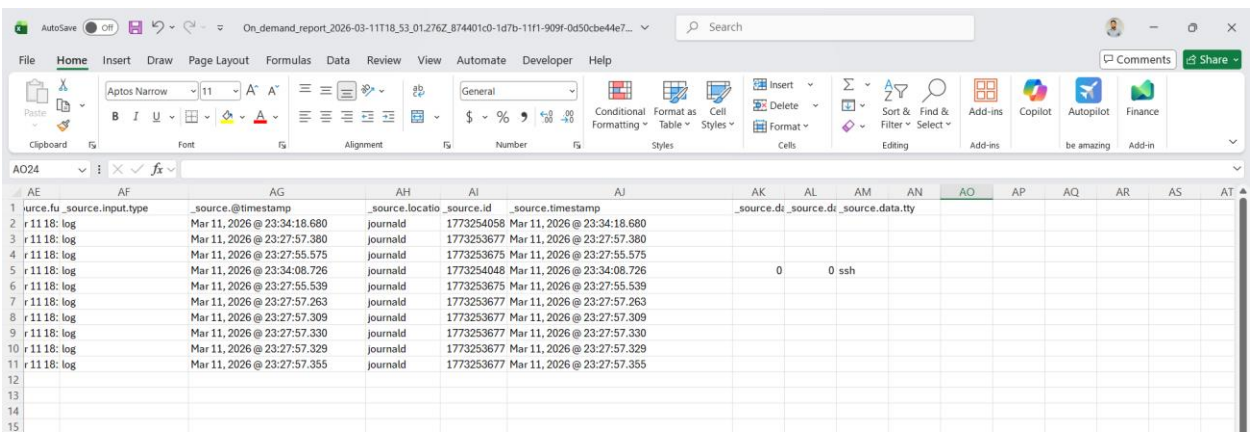
- The CSV export of your critical events. The file should be clean enough that someone could open it and understand what they are looking at without additional context.



_index	_id	_version	_score	_source.pr	_source.pr	_source.pr	_source.ag	_source.ag	_source.ag	_source.m	_source.da	_source.da	_source.ru	_source.ru	_source.ru	_source.ru	_source.ru	_source.ru	_source.ru	_source.ru	_source.ru	_source.ru	_source.ru	_source.ru	_source.ru	_s
wazuh-ale	EW4t3pwB	1	1	kali	sshd-sessi	########	192.168.5	Kali-Linux	2	Ubnutu-Se	192.168.5	mr-noob	FALSE	10	["10.2.4","	["164.312.	["CC6.1","(	syslog: Use	["syslog",":	["AU.14","/	["IV_35.7.c	1	["E			
wazuh-ale	vG4n3pwB	1	1	kali	systemd	########	192.168.5	Kali-Linux	2	Ubnutu-Se			FALSE	5				Systemd: S	["local","sy		["IV_35.7.c	8				
wazuh-ale	tG4n3pwB	1	1	kali	systemd	########	192.168.5	Kali-Linux	2	Ubnutu-Se			FALSE	5				Systemd: S	["local","sy		["IV_35.7.c	2				
wazuh-ale	Cm4t3pwE	1	1	kali	sshd-sessi	########	192.168.5	Kali-Linux	2	Ubnutu-Se	192.168.5	mr-noob	FALSE	5	["10.2.4","	["164.312.	["CC6.1","(	PAM: User	["pam","sy	["AU.14","/	["IV_35.7.c	1	["F			
wazuh-ale	s24n3pwB	1	1	kali	systemd	########	192.168.5	Kali-Linux	2	Ubnutu-Se			FALSE	5				Systemd: S	["local","sy		["IV_35.7.c	1				
wazuh-ale	t24n3pwB	1	1	kali	systemd	########	192.168.5	Kali-Linux	2	Ubnutu-Se			FALSE	5				Systemd: S	["local","sy		["IV_35.7.c	3				
wazuh-ale	uG4n3pwE	1	1	kali	systemd	########	192.168.5	Kali-Linux	2	Ubnutu-Se			FALSE	5				Systemd: S	["local","sy		["IV_35.7.c	4				
wazuh-ale	uW4n3pwE	1	1	kali	systemd	########	192.168.5	Kali-Linux	2	Ubnutu-Se			FALSE	5				Systemd: S	["local","sy		["IV_35.7.c	5				
wazuh-ale	um4n3pwE	1	1	kali	systemd	########	192.168.5	Kali-Linux	2	Ubnutu-Se			FALSE	5				Systemd: S	["local","sy		["IV_35.7.c	6				
wazuh-ale	u24n3pwB	1	1	kali	systemd	########	192.168.5	Kali-Linux	2	Ubnutu-Se			FALSE	5				Systemd: S	["local","sy		["IV_35.7.c	7				

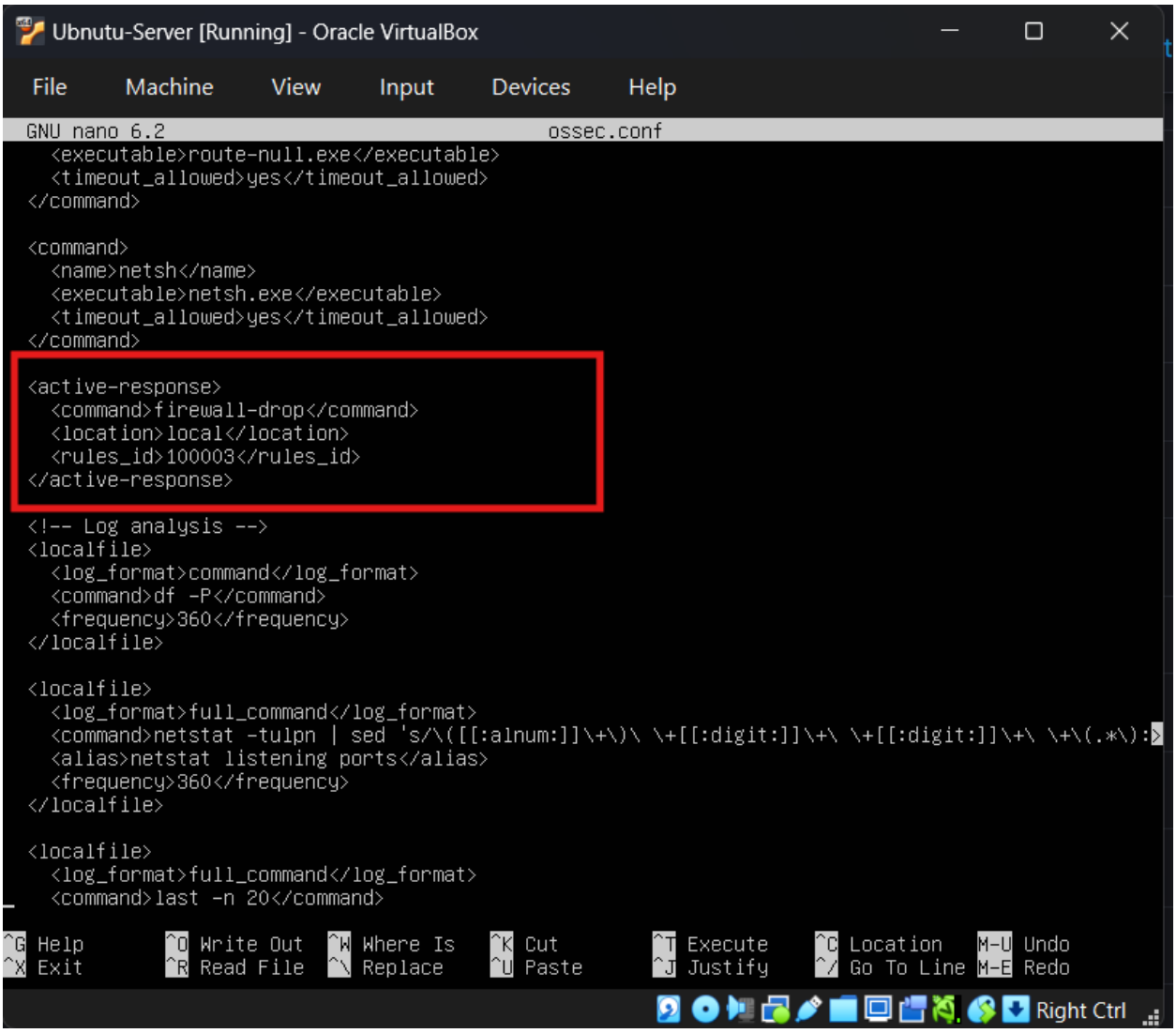
- A timeline of suspicious activity. List every event from this week that you would consider worth flagging in a real environment, in chronological order. For each one include the timestamp, the event description, the agent, and a one-sentence note on why you flagged it. This is the starting point for what would become a formal incident report in a real SOC workflow.



AE	AF	AG	AH	AI	AJ	AK	AL	AM	AN	AO	AP	AQ	AR	AS	AT
urce.fu	_source.input.type	_source.@timestamp	_source.locatio	_source.id	_source.timestamp	_source.da	_source.da	_source.data.tty							
r 11 18:	log	Mar 11, 2026 @ 23:34:18.680	journald	1773254058	Mar 11, 2026 @ 23:34:18.680										
r 11 18:	log	Mar 11, 2026 @ 23:27:57.380	journald	1773253677	Mar 11, 2026 @ 23:27:57.380										
r 11 18:	log	Mar 11, 2026 @ 23:27:55.575	journald	1773253675	Mar 11, 2026 @ 23:27:55.575										
r 11 18:	log	Mar 11, 2026 @ 23:34:08.726	journald	1773254048	Mar 11, 2026 @ 23:34:08.726	0	0	ssh							
r 11 18:	log	Mar 11, 2026 @ 23:27:55.539	journald	1773253675	Mar 11, 2026 @ 23:27:55.539										
r 11 18:	log	Mar 11, 2026 @ 23:27:57.263	journald	1773253677	Mar 11, 2026 @ 23:27:57.263										
r 11 18:	log	Mar 11, 2026 @ 23:27:57.309	journald	1773253677	Mar 11, 2026 @ 23:27:57.309										
r 11 18:	log	Mar 11, 2026 @ 23:27:57.330	journald	1773253677	Mar 11, 2026 @ 23:27:57.330										
r 11 18:	log	Mar 11, 2026 @ 23:27:57.329	journald	1773253677	Mar 11, 2026 @ 23:27:57.329										
r 11 18:	log	Mar 11, 2026 @ 23:27:57.355	journald	1773253677	Mar 11, 2026 @ 23:27:57.355										

Day 14 — Wazuh Active Response

- A document covering the full active response configuration and test workflow. This should read as a complete description of what you configured, why each setting is set the way it is, what you tested, and what the result was. Include the specific ossec.conf active-response block you configured.

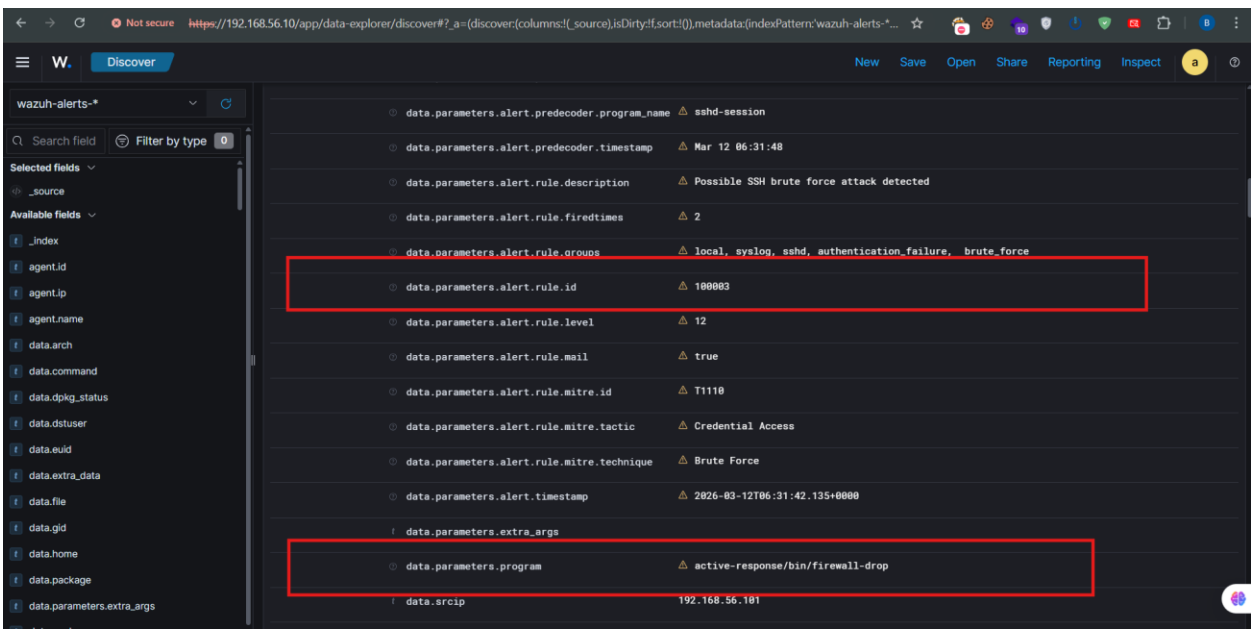


- A screenshot showing the blocked IP in iptables on your Ubuntu Server, with the Kali Linux IP clearly visible in the rule.

REPORT WEEK 2

```
┌──(mr-noob㉿kali)-[~]
└─$ sudo iptables -L -n
Chain INPUT (policy ACCEPT)
target     prot opt source               destination
DROP       all  --  192.168.56.101       0.0.0.0/0
DROP       all  --  192.168.56.101       0.0.0.0/0

Chain FORWARD (policy DROP)
target     prot opt source               destination
DROP       all  --  192.168.56.101       0.0.0.0/0
DROP       all  --  192.168.56.101       0.0.0.0/0
DOCKER-USER  all  --  0.0.0.0/0            0.0.0.0/0
DOCKER-ISOLATION-STAGE-1  all  --  0.0.0.0/0            0.0.0.0/0
ACCEPT     all  --  0.0.0.0/0            0.0.0.0/0            ctstate RELATED,ESTABLISHED
DOCKER     all  --  0.0.0.0/0            0.0.0.0/0
ACCEPT     all  --  0.0.0.0/0            0.0.0.0/0
ACCEPT     all  --  0.0.0.0/0            0.0.0.0/0

Chain OUTPUT (policy ACCEPT)
target     prot opt source               destination

Chain DOCKER (1 references)
target     prot opt source               destination

Chain DOCKER-ISOLATION-STAGE-1 (1 references)
target     prot opt source               destination
DOCKER-ISOLATION-STAGE-2  all  --  0.0.0.0/0            0.0.0.0/0
RETURN     all  --  0.0.0.0/0            0.0.0.0/0

Chain DOCKER-ISOLATION-STAGE-2 (1 references)
target     prot opt source               destination
DROP       all  --  0.0.0.0/0            0.0.0.0/0
RETURN     all  --  0.0.0.0/0            0.0.0.0/0

Chain DOCKER-USER (1 references)
target     prot opt source               destination
RETURN     all  --  0.0.0.0/0            0.0.0.0/0
```

- Screenshots of the active response alert in the Wazuh dashboard, showing the triggered event and the response action.



- **A short note, no more than half a page, on the risks of active response in a real environment and what safeguards you would put in place before enabling it on a production system. Think about what happens if a legitimate user fails five login attempts in a row, or if an attacker uses IP spoofing to make it look like a critical server is the source of the brute force. This is the kind of thinking that separates someone who can configure a tool from someone who understands how to use it safely.**

Active response can be very powerful but also risky in production environments. If configured incorrectly, it may block legitimate users who accidentally fail authentication several times. It may also cause service disruption if critical servers are mistakenly blocked due to misidentified traffic. Attackers may even attempt to exploit active response mechanisms by spoofing IP addresses to cause defensive systems to block legitimate infrastructure. To mitigate these risks, safeguards such as higher alert thresholds, whitelisting trusted IP addresses, and limiting automated responses to low-risk environments should be implemented. Careful monitoring and testing should also be performed before deploying active response in production systems.